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A cardiovascular disease prediction method based on cross-combination strategy and dynamic weighted stacking ensemble

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Cardiovascular disease (CVD) is one of the leading causes of death worldwide, early and accurate prediction is crucial for reducing both incidence and mortality rates. Aiming at the problems of insufficient feature association mining and poor generalisation ability of statically weighted ensemble in existing models, this paper proposes a CVD prediction method based on the ensemble of cross-combination strategy and dynamic weighted stacking ensemble (CCS-DWSE). The method firstly constructs a heterogeneous base model through a full combination of multi-feature selection techniques and multi-paradigm classifiers. Next, it designs a dynamic weighted stacking ensemble framework that adjusts the base model weights in real time, and adaptively fuses the prediction results of the K-Nearest Neighbors (KNN) meta-model. Finally, SHAP-based interpretability analysis is employed to quantify the contribution of each feature. The experimental results show that CCS-DWSE achieves accuracies of 98.32%, 92.33%, and 82.10% on three public datasets, with an area under the curve (AUC) of up to 0.9992, representing a 2.76% improvement over the best existing models. This study provides an efficient and scalable solution for CVD prediction and offers a novel perspective for collaborative multi-model learning on complex medical data.

Keywords Cardiovascular disease, Cross-combination strategy, Dynamic weighted stacking ensemble, Prediction method, SHAP-based interpretability

Cardiovascular disease (CVD) is one of the leading causes of death worldwide[1]. According to the World Health Organization (WHO), approximately 17.9 million people die from CVD each year, accounting for 32% of total deaths[2]. In China, the CVD mortality rate exceeds 40%, posing a significant threat to national health[3]. CVD is insidious and progresses rapidly. Although traditional diagnostic techniques are commonly used, they are difficult to detect in the early stages. These techniques suffer from insufficient sensitivity and rely on doctors' experience, which may cause patients to miss the optimal timing for intervention. Therefore, there is an urgent need for an efficient and accurate CVD prediction method to enable early intervention, which is crucial for reducing incidence and mortality rates.

In recent years, machine learning-based CVD prediction methods have rapidly developed in academia and clinical applications[4], and can be roughly divided into three categories, but all have certain limitations: (i) Traditional statistical models (e.g. the Framingham Risk Score[5] and SCORE model[6] rely on linear regression assumptions. This simplifies the correlation between risk factors and fails to capture the complex, nonlinear relationships between metrics such as cholesterol and blood sugar. (ii) Although single machine learning models (e.g. Random Forest[7] and XGBoost[8] perform well on specific datasets, they are sensitive to data distribution and their performance decreases significantly when transferred to a general population. Furthermore, the "black box" characteristic restricts clinical interpretability. (iii) Ensemble learning models (e.g. adaptive modular neural network, AMNN[9] and one-dimensional convolutional neural network, 1D-CNN[10] improve efficiency by modularising and simplifying calculations, or by automatically extracting features. However, they may miss cross-group feature association or require a large amount of labelled data, which is limited in small sample scenarios. Among the existing ensemble methods, some adopt a static weighting strategy where the weights are calculated based on model performance but remain unchanged once determined. For instance, MCF[11] assigns

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weights to base models by conducting a comprehensive score evaluation based on the multi-criteria decision-making method, which makes it difficult to adapt to the dynamic changes of clinical data. Another category of methods employs dynamic weighting yet is mostly limited to homogeneous base models, failing to fuse the complementary advantages of heterogeneous models for full exploration of data diversity.

To address the above issues, this paper proposes a CVD prediction method based on CCS-DWSE. The main contributions are as follows:

1) To address the problem of insufficient feature correlation mining, a cross-combination strategy (CCS) is proposed. This strategy aims to combine multiple feature selection techniques and multi-paradigm classifiers to screen key features from multiple dimensions.
2) To address the problem of insufficient generalisation ability of static weighted collection, a dynamic weighted stacking ensemble (DWSE) framework is designed. This framework adjusts the weights of the base models in real time and fuses their prediction results adaptively through a KNN meta-model. This improves the model's robustness under changes in data distribution.
3) Combined with SHAP (Shapley additive explanations) interpretability analysis, the results indicate that smoking, blood lipids and blood glucose are the key influencing factors for the model to predict CVD. This provides a reference for understanding model decisions and feature selection in subsequent related studies.

Related works

Feature selection applications in medical data

Feature selection is a crucial pre-processing step for improving the performance of medical data modelling. Previous studies have mainly employed single strategy. For example, the chi-square test based on statistical tests relies on linear assumptions[12,13], making it difficult to capture complex non-linear relationships in medical data. Model-driven L1 regularization dynamically optimizes feature subsets during training[14,15], but tends to fall into local optimal solutions. Recent hybrid methods, such as ReliefF combined with the Rough Set approach (RFRS)[16], improve robustness through multi-strategy fusion. However, it is limited in feature interaction analysis. Existing methods have two shortcomings: filter methods ignore the interaction between features and the model during optimisation. Embedded methods overly depend on model-specific feature importance judgments. To address these issues, this paper uses a combination of multi-feature selection techniques (filter, wrapper and embedded) to select features based on statistical significance, model performance and data distribution.

Advances in ensemble learning in disease prediction

Ensemble learning improves generalization ability by combining multiple base models, its effectiveness depends on model diversity. Traditional bagging and boosting methods use homogeneous base models[17], which limits feature extraction capability; in contrast, stacking improves performance by employing heterogeneous models[18]. In existing studies, Yuan[19] proposed a GBDT algorithm based on fuzzy theory and bagging to improve the performance, but its static weight allocation mechanism struggles to adapt to changes in data distribution. Mienye[20] proposed an artificial neural network (ANN) method based on sparse autoencoders to extract features and improve performance, but it is computationally complex. Mahesh[21] proposed a hybrid ensemble learning method combining Bayesian networks with radial basis functions to support diabetes diagnosis and treatment decisions. However, the performance of this model is dependent on the dataset used. In addition, Abdelhamid[22] proposes an ensemble framework fusing convolutional neural networks and vision transformers for the medical image classification of pulmonary embolism, which effectively improves detection accuracy and verifies the effectiveness of ensemble learning in the prediction of cardiovascular disease imaging data. Existing ensemble learning models have two limitations: one is insufficient model diversity due to the use of identical base models. The other is difficulty in adapting to imbalanced datasets due to the adoption of static weighted ensembles.

Materials and methods

Among existing ensemble methods, MCF assigns weights to base models through comprehensive scoring based on multi-criteria decision-making, with its weight calculation relying on the fixed criteria matrix and equal-weight strategy. This method can effectively synthesize the multi-dimensional performance of models in static scenarios, but once the weights are determined, they are no longer adjusted, leading to limited adaptability when facing changes in the distribution of clinical data. Drawing on the cross-combination idea in MCF, this paper introduces a dynamic weighting and error feedback iteration mechanism. We propose a DWSE framework. This method enables adaptive updates of weights according to the real-time performance of models, thereby better adapting to fluctuations in data distribution and improving generalization ability.

CCS fusion

To address the inadequacy of existing methods in mining feature associations, this paper proposes a CCS based on the full combination of multiple feature selection methods and multi-paradigm classifiers. The CCS method aims to capture feature interactions as comprehensively as possible through a diversity-driven strategy. It constructs 112 candidate base models by fully combining 8 feature selection methods with 14 multi-paradigm classifiers, excavating feature associations from diverse dimensions such as statistical significance, model-driven relevance and data distribution to capture relevant features as comprehensively as possible. The base models include those with strong capabilities in mining nonlinear relationships, such as XGBoost, MLP, and Extra Trees, which can effectively reduce the risk of missing high-dimensional nonlinear dependencies by individual base models. Additionally, combined with a dynamic weighted fusion mechanism, the ensemble model can further integrate the complementary advantages of different base models, partially compensating for the insufficient capture of nonlinear patterns by single base models. Ref.11 used ovarian cancer data, while this paper focuses

Number	Method name	References
1	Tree-based Feature Selection	Pal & Parija[7].
2	ANOVA F-value Selection (F score)	Cai et al.[11]
3	Chi-Squared	Fu et al.[12]
4	Mutual Information	Haochun Qi[13].
5	Variance Threshold	Shekar & Dagnew[15].
6	L1-based Feature Selection	
7	Recursive Feature Elimination (RFE)	Gao et al.[23]
8	Gini Index	Alessio Guandalini[24].

Table 1. The 8 feature selection methods employed for the base models.

Number	Classifier name	References
1	Naive Bayes	HUANG et al.[4]
2	Logistic Regression (LR)	Mahesh et al.[21]
3	Random Forest (RF)	
4	Decision Tree	
5	Support Vector Machine (SVM)	
6	K-Nearest Neighbors (KNN)	
7	Extremely Randomized Trees (Extra Trees)	Nzenwata et al.[25]
8	Adaptive Boosting (AdaBoost)	Sultan Bin Habib et al.[26]
9	Categorical Boosting (CatBoost)	
10	eXtremeGradient Boosting (XGBoost)	
11	Gradient Boosting Machine (Gradient Boosting)	
12	Light Gradient Boosting Machine (LGBM)	
13	Stochastic Gradient Descent (SGD	Dagal et al.[27]
14	Multiple-Layer Perceiver (MLP)	Arnob et al.[28]

Table 2. The 14 classifiers employed for the base models.

on heart-related indicators such as blood pressure, heart rate, and cholesterol. It places greater emphasis on both linear and non-linear relationships between these features and the disease. Therefore, this paper selects feature selection methods capable of handling both continuous and categorical data, while uncovering feature interactions (see Table 1). Additionally, it employs classifiers suited for weakly correlated data and complex pattern mining (see Table 2), thereby enhancing the accuracy and reliability of CVD prediction.

DWSE framework

Stacking is a common ensemble learning method. Its core idea is to improve overall prediction performance by combining the prediction results of multiple base models. First, multiple base models make predictions on the training data. Then, the predicted values of these models are treated as new features (i.e., meta-features) and input into a meta-model. Finally, the meta-model learns these meta-features and integrates the strengths of each base model to provide the final prediction result. The base models should be as diverse as possible to capture different patterns, while the meta-model can be any type of model.

However, traditional stacking ensembles typically combine the predictions of base models using statical weights. This fixed-weight strategy exhibits poor adaptability to changes in data distribution, leading to insufficient model generalisation. To address this issue, this paper proposes a DWSE framework, as shown in Fig. 1,which comprises three key steps.

(1) Dynamic weight allocation. First, assign equal initial weights to each base model. Then, using a predefined error feedback mechanism, iteratively update the weights based on the prediction accuracy error during k-fold cross-validation, ultimately obtaining the optimal weights for each base model.
(2) Meta-feature generation. Use the optimal dynamic weights obtained from the above steps to weight and fuse the prediction probability outputs of the base model, generating meta-feature dataset for training the meta-model.
(3) Meta-model training and prediction. Train the KNN meta-model on the meta-feature dataset after dynamic weighted fusion and perform the final prediction.

Through the aforementioned dynamic weight allocation mechanism, this framework dynamically adjusts the weights of its base models based on their real-time performance, significantly enhancing the ensemble model's adaptability and generalisation capability when confronted with complex and variable data.

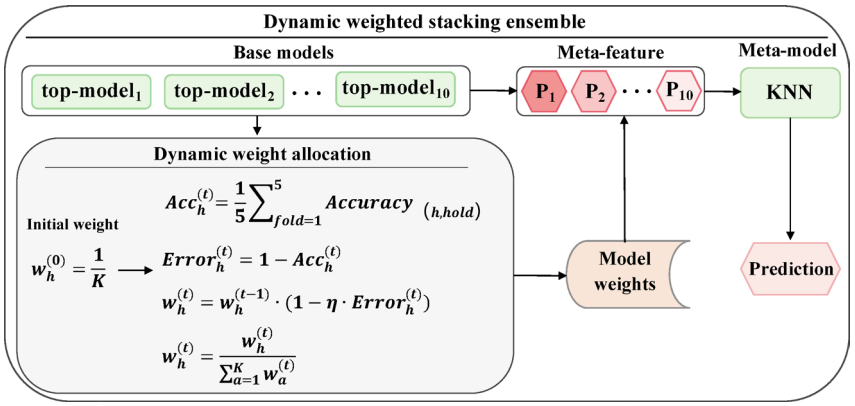


Fig. 1. DWSE framework.

Input: Base models: $H = \{H_1, H_2, \ldots, H_k\}$; predicted probabilities of H_h on the validation set: P_h; true labels of the validation set: y_{val}; iteration candidates: $T_{\text{grid}} = [5, 10, 15]$; learning -rate candidates: $\eta_{\text{grid}} = [0.0001, 0.001, 0.01]$.
Output: Optimal weight vector: $w^* = (w_1^*, w_2^*, \ldots, w_K^*)$; Optimal iterations: T^*; optimal learning rate: η^*.

$P \leftarrow []$ // store all P_h
for each H_h in H **do**
  read P_h from memory
  append P_h to P
end for
$best_score \leftarrow 0$
for T in T_{grid} **do**
  for η in η_{grid} **do**
    for $h = 1 \ldots K$ **do**
      $w_h^{(0)} \leftarrow 1/K$ // equal initial weights
    end for
    for $t = 1 \ldots T$ **do**
      for $h = 1 \ldots K$ **do**
        $Acc_h^{(t)} \leftarrow \frac{1}{5}\sum_{fold=1}^{5} Accuracy_{(h,fold)}$
        $Error_h^{(t)} \leftarrow 1 - Acc_h^{(t)}$
        $w_h^{(t)} \leftarrow w_h^{(t-1)} \cdot \left(1 - \eta \cdot Error_h^{(t)}\right)$
        if $\sum_{h=1}^{K} \left| w_h^{(t)} - w_h^{(t-1)} \right| < \epsilon$ **then**
          break
        end if
      end for
      $w_h^{(t)} \leftarrow \frac{w_h^{(t)}}{\sum_{a=1}^{K} w_a^{(t)}}$
    end for
    $\hat{P} \leftarrow \sum_{h=1}^{K} w_h^* \cdot P_h$
    score ← accuracy (P, y_{val})
    if score > $best_score$ **then**
      $best_score \leftarrow$ score
      $w^* \leftarrow w$
      $T^* \leftarrow T$
      $\eta^* \leftarrow \eta$
    end if
  end for
end for

Algorithm 1. DWSE.

Weighting update mechanism
The dataset is divided into the training set and testing set according to the ratio of 7:3, with a 5-fold cross-validation strategy employed in the training set. The dynamic weight update mechanism performs feedback-based iterative adjustments based on the real-time prediction errors of the base models across each validation fold, enabling the ensemble weights to adapt to the distribution fluctuations of the internal divisions of the training data and obtain a relatively adaptive weight combination for prediction on the testing set. The specific weight update process of the DWSE framework is as follows.

First, the average accuracy error of each base model is calculated in each iteration, with the expression given by:

$$Error_h^{(t)} = 1 - Acc_h^{(t)} \tag{1}$$

Where $Acc_h^{(t)}$ denotes the 5-fold cross-validation average accuracy of the h-th base model at iteration t, calculated as follows:

$$Acc_h^{(t)} = \frac{1}{5}\sum\nolimits_{fold=1}^{5} Accuracy_{(h,fold)} \tag{2}$$

Based on this error, the weights are updated according to the following feedback rule:

$$w_h^{(t)} = w_h^{(t-1)} \cdot \left(1 - \eta \cdot Error_h^{(t)}\right) \tag{3}$$

In the formula, $w_h^{(t)}$ denotes the weight of the h-th base model after the t-th iteration. The initial weight $w_h^{(0)} = 1/\mathrm{K}$ (where $K = 10$ in this study, representing the number of base models after $Top - K$ selection). η denotes the learning rate, which is used to control the adjustment amplitude of global weights. Its value needs to strike a balance between convergence speed and update stability. Preliminary experimental observations show that when $\eta < 0.0001$, the number of iterations required for model convergence increases, resulting in low training efficiency. When $\eta > 0.01$, the weight update process is prone to fluctuations. Therefore, the search range of η is set to [0.0001, 0.001, 0.01]. Further verification through sensitivity analysis on this basis indicates that the fluctuation range of the model's AUC within this range is less than 0.5%, which verifies the rationality of η selection and the robustness of the model.

After each round of weight updates, normalization is performed to ensure the sum of all base model weights equals 1. The normalization formula is:

$$w_h^{(t)} = \frac{w_h^{(t)}}{\sum\nolimits_{a=1}^{K} w_a^{(t)}} \tag{4}$$

The iteration termination condition is: either reaching the preset maximum iteration count T, or the sum of absolute changes in all weights between two consecutive iterations falling below the threshold $\epsilon = 10^{-5}$:

$$\sum\nolimits_{h=1}^{K} \left| w_h^{(t)} - w_h^{(t-1)} \right| < \epsilon \tag{5}$$

When any of the above conditions is satisfied, the iterative process terminates. Upon termination, the optimal weights for each base model are output for use in the subsequent prediction stage's weighted fusion.

DWSE algorithm
The DWSE algorithm is as follows.

CCS-DWSE model

The CCS-DWSE model achieves end-to-end optimization for CVD prediction tasks through algorithmic-level collaborative interaction across three-tier modules. Each module functions independently while exchanging data and results via interfaces, forming an integrated whole. The overall architecture is illustrated in Fig. 2, with the specific construction as follows.

(1) Feature selection and model layer. When performing feature selection based on CCS, all operations are executed within a five-fold cross-validation loop on the training set. Within each cross-validation fold, only the current training set is utilized to adapt eight feature selection methods (see Table 1, including filtering, wrapping, and embedding approaches) to generate multiple feature subsets from different perspectives. Each feature subset is then fully combined with 14 multi-paradigm classifiers (see Table 2) to yield 112 candidate base models. Following 5-fold cross-validation and AUC ranking, the top 10 models were selected to form the set $H = \{H_1, H_2, ..., H_{10}\}$. Their prediction probability list $P = \{P_1, P_2, ...P_{10}\}$ on the training set was output as input for the subsequent layer of stacking ensemble layer.
(2) Stacking ensemble layer. Adaptively fuse prediction probabilities based on the DWSE framework. Calculate the optimal weight $w_h{}^*$ for each base model using the dynamic weight update mechanism described in

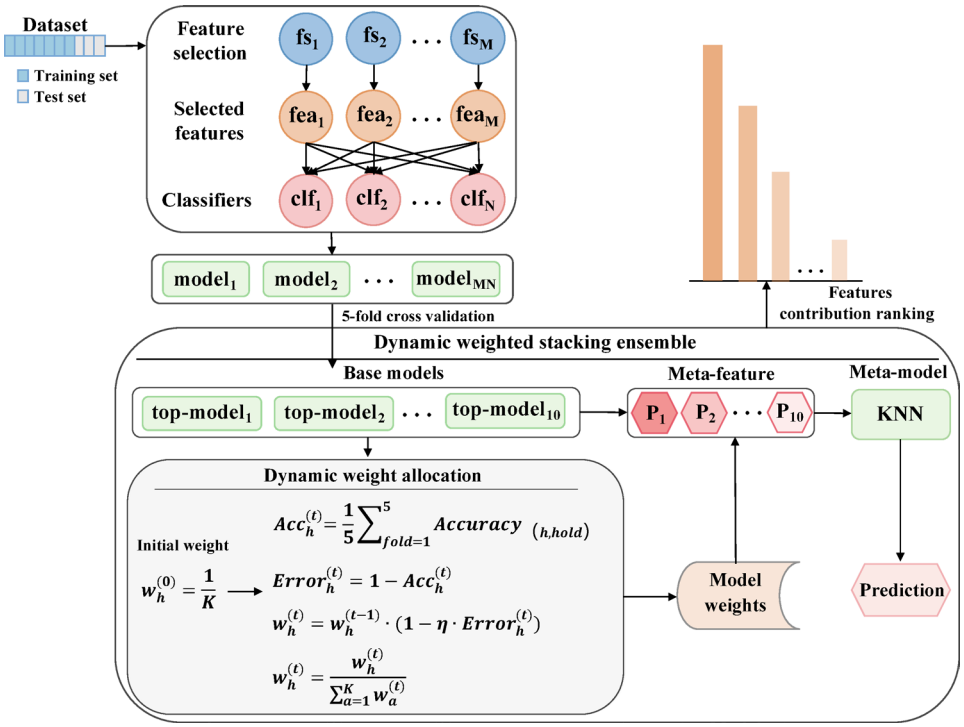


Fig. 2. CCS-DWSE model.

Input: Training set: D_{train} ; testing set: D_{test}
Output: Predicted class labels for D_{test} : ŷ; metrics: {accuracy, precision, recall, F1 -score, AUC}; SHAP-based feature contribution diagram: $feature_{contrib}$
1: split D_{train} and D_{test} into feature matrix X and target vector y
2: **for** each $fs \in FS_M$ **do** // FS_M = {list of feature-selection methods}
3: apply fs to X_{train} → X_{fs}
4: **end for**
5: **for** each $clf \in CL_N$ **do** // CL_N = {list of candidate classifiers}
6: initializeclf with default hyper-parameters
7: **end for**
8: **for** each $fs \in FS_M$ **do**
9: **for** each $clf \in CL_N$ **do**
10: X_{fs} = features selected by fs
11: $train_{clf} = clone(clf)$
12: $train_{clf}$. fit$(X_{fs_{train}}, y_{train})$
13: $cv_{auc} = cross_{val_{score}}(train_{clf}, X_{fs_{train}}, y_{train}, cv = 5, scoring = 'auc')$. mean()
14: store$(fs, clf, cv_{auc}, model_{object}, X_{fs})$
15: **end for**
16: **end for**
17: Rank all (fs, clf) pairs by cv_{auc} descending
18: Select top-K pairs→ $H = \{H_1, H_2, \ldots, H_k\}$
19: **for** each $H_k \in H$ **do**
20: save predicted probabilities on D_{test} : P_k
21: **end for**
22: Run **Error! Reference source not found,** obtain optimal weights w*, final fusion P, and metrics
23: Fit SHAP explainer on the weighted ensemble
24: Compute SHAP values on D_{test}
25: Generate and return $feature_{cont\,rib}$ diagram

Algorithm 2. CCS-DWSE.

Algorithm 1. Generate the weighted feature vector $\widehat{P} = \sum_{h=1}^{K} w_h^* \cdot P_h$ for the entity, input it into the KNN meta-model to obtain the predicted label $\widehat{y}$ and probability.

(3) SHAP-based interpretability analysis layer. Conduct interpretability analysis on the predicted results from the upper layer. This layer employs the Kernel SHAP algorithm to quantify the contribution of each input feature to the prediction outcome for a single sample (i.e., Shapley values). By aggregating the Shapley values across all test samples, a feature global importance ranking diagram is generated (as shown in Figs. 10–12), thereby enhancing the model's transparency and credibility within medical decision-making scenarios.

The CCS-DWSE algorithm is as follows.

Complexity analysis

The computational cost of CCS-DWSE primarily concentrates across three stages: feature selection, base model training, and dynamic weighted stacking. Assuming the dataset comprises N samples and F raw features, with M feature selection methods employed to construct K base models, the time complexity analysis for each stage is as follows:

(1) Feature selection phase. The computational complexity of feature selection depends on the method type: filtering methods (such as variance selection or mutual information) exhibit a time complexity of $O(F \cdot N \cdot M)$. Wrapping methods (such as recursive feature elimination) exhibit a complexity of $O(F \cdot N \cdot M^2)$. As this study incorporates both categories of methods, the overall complexity for this stage approximates $O(F \cdot N \cdot M^2)$.

(2) Base model training phase. The time complexity for training K base models depends on the classifier type: for tree models, the complexity of training a single model is $O(N \cdot D \cdot logN)$, where D is the tree depth. Thus, the total complexity for K models is $O(K \cdot N \cdot D \cdot logN)$. For linear models (e.g., LR), the complexity is $O(K \cdot N \cdot F)$. In practice, base models can be trained in parallel, so the complexity for this phase is the maximum value among the two model types: $O(K \cdot N \cdot max(D \cdot logN, F))$.

(3) Dynamic weighted stacking phase. Dynamic weight optimization requires T iterations across K base models (each iteration comprising 5-fold cross-validation, accuracy calculation, and weight update): the core computational complexity of cross-validation and accuracy calculation is $O(K \cdot N \cdot F)$, whilst weight updates incur negligible constant-order overhead. Therefore, the total complexity for T iterations is $O(T \cdot K \cdot N \cdot F)$.

In summary, the overall time complexity of CCS-DWSE is: $O(F \cdot N \cdot M^2 + K \cdot N \cdot (max(D \cdot logN, F) + T \cdot F))$.

Results and discussion

Experimental setup

To ensure the of experimental results and effectively mitigate risks of overfitting and information leakage, each dataset was randomly partitioned into the training set (70%) and testing set (30%) according to its original category distribution. The testing set remained entirely independent throughout the process, serving solely for final performance evaluation without participating in any parameter optimization or data fitting during the modelling stages.

To address the class imbalance issue in the Framingham dataset (with a post-preprocessing positive-to-negative sample ratio of approximately 7.07:1), during the base model training phase, the "class_weight = 'balanced'" parameter was enabled for models supporting category weight adjustment—such as XGBoost, RF, and LR. This allows them to automatically adjust sample weights based on the training set's class distribution, mitigating the dominant influence of majority-class samples on model training. During the ensemble optimization phase, as the aforementioned category-weight-supporting models were still employed in dynamic weighted stacking ensemble, the category weight adjustment strategy was similarly maintained to enhance the balanced classification decision-making of the ensemble model.

Data sources

This paper selects three public datasets for research, namely the Cleveland Heart Disease Dataset[29] provided by the University of California Irvine (UCI) database, the Framingham Heart Disease Dataset[30] obtained from the Kaggle platform and the CVD Dataset[31]. Basic information for each dataset is presented in Table 3.

Dataset	Sample size	Number of features	Feature Types	Variable	Data characteristics
Cleveland	303	14	Numerical: 6 Categorical: 8	age, sex, cp., trestbps, chol, fbs, restecg, thalach, exang, oldpeak, slope, ca., thal, target	Balanced, small-scale
Framingham	4240	16	Numerical: 8 Categorical: 8	male, age, education, currentSmoker, cigsPerDay, BPMeds, prevalentStroke, prevalentHyp, diabetes, totChol, sysBP, diaBP, BMI, heartrate, glucose, TenYearCHD	Highly imbalanced
CVD	70,000	12	Numerical: 5 Categorical: 7	age, gender, height, weight, ap_hi, ap_lo, cholesterol, gluc, smoke, alco, active, cardio	Large-scale, features containing one-hot encoding

Table 3. Dataset description.

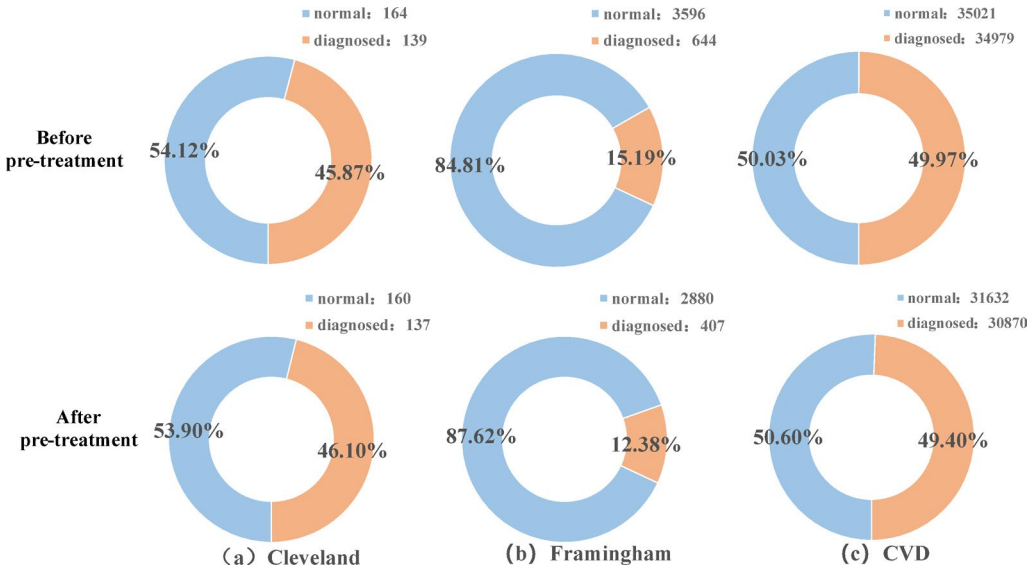


Fig. 3. Distribution of target variables before and after preprocessing.

Statistic	Before interpolation (missing samples)	After interpolation (all samples)	Difference rate (%)
Mean	81.964	81.580	0.462
Standard Deviation	23.951	17.164	28.336
Median	78.000	80.000	1.564
Interquartile range Q1	71.000	72.000	1.408
Interquartile range Q3	87.000	87.250	0.287

Table 4. Comparison of distribution characteristics before and after glucose variable interpolation.

Data pre-processing

For the Cleveland dataset, the values 3, 6, and 7 in the "thal" variable were recoded as 0, 1, and 2 respectively, and the original target variable was converted into a binary label (0: normal, 1: diagnosed). The proportion of missing values is below 2%. Excluding missing samples has negligible impact on the overall data. To preserve data authenticity and distributional consistency, six records containing missing values were deleted. In the Framingham dataset, the "glucose" variable exhibits approximately 10% missing data. As a key clinical indicator for cardiovascular disease diagnosis, this variable holds considerable value for model prediction accuracy. Direct deletion of relevant samples could result in substantial loss of valid information and compromise the integrity of the data distribution. Consequently, the KNNImputer method was employed to impute this variable. For variables with a missing rate below 3%, records containing missing values were directly deleted. Concurrently, the interquartile range (IQR) method was applied to identify and remove outliers from each variable, thereby enhancing data quality. The CVD dataset has no missing values, only abnormal samples were removed to preserve the features.

Figure 3 depicts the distribution of the target variable for the three datasets before and after preprocessing, respectively. The Cleveland dataset contained 303 samples before preprocessing and 297 samples after preprocessing. The Framingham dataset had 4,240 samples before preprocessing and 3,287 samples after preprocessing. The CVD dataset included 70,000 samples before preprocessing and 62,502 samples after preprocessing.

To verify whether the KNNImputer method introduces significant bias to the "glucose" variable in the Framingham dataset, a quantitative analysis of distribution consistency was conducted using the Kolmogorov-Smirnov (K-S) test, supplemented by statistical feature comparisons. The test employed the original non-missing "glucose" samples as the reference group, with the entire KNNImputer-interpolated dataset serving as the validation group. A significance level of $\alpha = 0.05$ was set, with the null hypothesis stating no significant difference in the "glucose" distribution between the two groups. The K-S test results showed a test statistic $D = 0.0569$ and a p-value of 0.1963 for the "glucose" variable before and after interpolation. As $p > 0.05$, the null hypothesis cannot be rejected. Consequently, the distribution of "glucose" in the interpolated full dataset remains fundamentally consistent with that of the original non-missing dataset, indicating no significant bias introduction.

Combined with the statistical feature comparison results in Table 4 Comparison of distribution characteristics before and after glucose variable interpolation, the changes in the mean, median, and quartiles (Q1, Q3) of the "glucose" variable after KNNImputer interpolation were all controlled within a relatively small range (difference rate $\leq 5\%$), demonstrating that the interpolation method possesses a certain capacity to preserve the overall

Label	Predict confirmed	Predict normal
Actual confirmed	TP	FN
Actual normal	FP	TN

Table 5. The relationship of the confusion matrix.

K	Average AUC	Prediction time (seconds)
5	0.92 ± 0.03	942.17
10	$\mathbf{0.95 \pm 0.02}$	**1242.67**
15	0.89 ± 0.02	1512.42
20	0.82 ± 0.04	2032.02

Table 6. Performance and computational time comparison across different base model sizes.

structure of the original data. The variation in standard deviation primarily stems from the imputation of originally missing samples. All imputed values fall within the clinically common fasting blood glucose range, with no apparent outliers observed.

Evaluation metrics

In medical diagnostics, the confusion matrix is a fundamental method for evaluating the performance of a classifier. For binary classification problems involving "positive class" and "negative class", the confusion matrix contains true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), as shown in Table 5. In this paper, the confirmed group is the "positive class" and the normal group is the "negative class".

Calculate the five-evaluation metrics, which include Accuracy, Precision, Recall, F1-score, and AUC (Area Under the ROC Curve). The formulas for these calculations are as follows.

Accuracy:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{6}$$

Precision:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{7}$$

Recall:

$$Recall = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{8}$$

F1-score:

$$\text{F1} - \text{score} = 2^{*}\frac{\text{Precision}^{*}\text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

AUC is a metric used to evaluate the performance of a classifier at different thresholds. AUC values range from 0 to 1, with values closer to 1 indicating better performance.

Selection of base model quantity

The number of base models (K) is a key parameter balancing the diversity and complexity of the ensemble model. Its value must follow the following trade-off principle: when K is too small, the number of models participating in the ensemble is limited, making it difficult to fully exploit the complementary advantages of heterogeneous models. When K is too large, a large number of base models with similar performance or redundancy will be introduced, increasing model complexity and the risk of overfitting. To determine the optimal K value, a sensitivity analysis was conducted through gradient control experiments. Four schemes with K = 5,10,15,20 were set up, and 5-fold cross-validation was adopted to evaluate the average AUC and single-round prediction time on three datasets. The results are presented in Table 6.

Results indicate that when $K = 10$, the model achieved the highest mean AUC (0.95 ± 0.02), with prediction time (1242.67 s) remaining within a reasonable range. Although $K = 5$ exhibited lower computational time (942.17 s), its AUC was only 0.92, indicating insufficient performance. For $K > 10$, the AUC declined significantly (to 0.89 at $K = 15$ and 0.82 at $K = 20$), while computational time increased substantially (reaching 2032.02 s at $K = 20$), revealing a pronounced performance inflection point. Therefore, $K = 10$ achieves the optimal balance between predictive accuracy and computational efficiency, and is thus determined as the final ensemble size.

Model	Parameter name	Value range	Accuracy	Precision	Recall	F1-score	AUC
Naive Bayes	var_smoothing	[**1e-9**, 1e-8, 1e-7, 1e-6, 1e-5]	0.9731	0.9778	0.9635	0.9706	0.9960
LR	C	[0.01, 0.1, 1, **10**]	0.9731	0.9778	0.9635	0.9706	0.9973
	penalty	['l1', **'l2'**, 'none']					
	solver	['lbfgs', **'liblinear'**, 'sag']					
MLP	activation	[**'tanh'**, 'relu', 'logistic']	0.9731	0.9778	0.9635	0.9706	0.9964
	alpha	[**0.0001**, 0.001, 0.01]					
	hidden_layer_sizes	[(50,), (100,), **(50, 50)**]					
SVM	C	[0.01, 0.1, 1, **10**]	0.9697	0.9776	0.9562	0.9668	0.9965
	kernel	[**'linear'**, 'poly', 'rbf']					
	gamma	[**'auto'**, 0.001, 0.01, 0.1]					
KNN	n_neighbors	[**3**, 5, 7]	**0.9832**	**0.9853**	**0.9781**	**0.9817**	**0.9992**
	p	[**1**, 2]					
SGD	loss	[**'log'**, 'modified_huber']	0.9764	0.9851	0.9635	0.9742	0.9938
	alpha	[0.0001,**0.001**, 0.01]					
	penalty	[**'l1'**, 'l2', 'none']					

Table 7. Prediction results of different meta-models on the Cleveland dataset. Note: bold values indicate the best results.

Model	Parameter name	Value range	Accuracy	Precision	Recall	F1-score	AUC
Naive Bayes	var_smoothing	[1e-9, 1e-8, 1e-7, 1e-6, **1e-5**]	0.8035	0.3124	0.4849	0.3812	0.7635
LR	C	[**0.01**, 0.1, 1, 10]	0.8923	0.6879	0.2383	0.3540	0.8145
	penalty	['l1', 'l2', **'none'**]					
	solver	['lbfgs', 'liblinear', **'sag'**]					
MLP	activation	['tanh', **'relu'**, 'logistic']	0.8932	0.6750	0.2654	0.3810	0.8960
	alpha	[0.0001, 0.001, **0.01**]					
	hidden_layer_sizes	[(50,), (100,), **(50, 50)**]					
SVM	C	[**0.01**, 0.1, 1, 10]	0.8762	0.6706	0.2562	0.3668	0.7916
	kernel	[**'linear'**, 'poly', 'rbf']					
	gamma	[**'auto'**, 0.001, 0.01, 0.1]					
KNN	n_neighbors	[**3**, 5, 7]	**0.9233**	**0.9039**	**0.5037**	**0.6193**	**0.9539**
	p	[**1**, 2]					
SGD	loss	['log', **'modified_huber'**]	0.8926	0.6552	0.2801	0.3924	0.8087
	alpha	[**0.0001**,0.001, 0.01]					
	penalty	[**'l1'**, 'l2', 'none']					

Table 8. Prediction results of different meta-models on the Framingham dataset.

Meta-model selection in DWSE framework

To determine the optimal meta-model, this paper first evaluated the classifiers listed in Table 2. For smaller datasets, complex and highly learnable models (such as AdaBoost, CatBoost, and XGBoost) are prone to overfitting, so these models were excluded. Then, the hyperparameters of relatively robust meta-models (including Naive Bayes, LR, MLP, SVM, KNN, and SGD) were optimised using grid search and five-fold cross-validation. The parameter ranges and prediction results for these models on the Cleveland, Framingham, and CVD datasets are shown in Tables 7, 8 and 9.

The results show that KNN achieved optimal performance across all three datasets: on the Cleveland dataset, it attained an AUC of 0.9992 with accuracy and recall of 98.32% and 97.81% respectively. On the Framingham dataset, it achieved an AUC of 0.9539 with accuracy of 92.33%. On the CVD dataset, it attained an AUC of 0.9028, with all metrics outperforming other models. The performance of KNN relies on the local distribution characteristics of data. Since the CVD dataset exhibits large-scale sparse features, tree-based models such as XGBoost and Gradient Boosting can directly adapt to this data structure through feature importance ranking, whereas using KNN alone struggles to fully capture the associations between high-dimensional sparse features, leading to inferior independent classification performance compared to tree-based models and thus suboptimal results at the base model prediction stage. However, in the CCS-DWSE framework, KNN serves as a meta-model rather than an independent base model, and its advantage lies not in directly classifying raw features but in adaptively fusing the prediction results of multiple base models without requiring pre-specified data distribution assumptions, thereby enabling CCS-DWSE to achieve favorable experimental results in the final prediction. Therefore, KNN was ultimately selected as the meta-model.

Model	Parameter name	Value range	Accuracy	Precision	Recall	F1-score	AUC
Naive Bayes	var_smoothing	[**1e-9**, 1e-8, 1e-7, 1e-6, 1e-5]	0.7355	0.7576	0.6831	0.7184	0.8037
LR	C	[**0.01**, 0.1, 1, 10]	0.7452	0.7601	0.7074	0.7328	0.8212
	penalty	['l1', 'l2', **'none'**]					
	solver	[**'lbfgs'**, 'liblinear', 'sag']					
MLP	activation	[**'tanh'**, 'relu', 'logistic']	0.7458	0.7740	0.6856	0.7271	0.8210
	alpha	[0.0001, **0.001**, 0.01]					
	hidden_layer_sizes	[**(50,)**, (100,), (50, 50)]					
SVM	C	[0.01, **0.1**, 1, 10]	0.7825	0.7891	0.7703	0.7796	0.8516
	kernel	['linear', 'poly', **'rbf'**]					
	gamma	[**'auto'**, 0.001, 0.01, 0.1]					
KNN	n_neighbors	[**3**, 5, 7]	**0.8210**	**0.8234**	**0.8119**	**0.8176**	**0.9028**
	p	[**1**, 2]					
SGD	loss	[**'log'**, 'modified_huber']	0.7613	0.7682	0.7495	0.7587	0.8325
	alpha	[0.0001,0.001, **0.01**]					
	penalty	[**'l1'**, 'l2', 'none']					

Table 9. Prediction results of different meta-models on the CVD dataset.

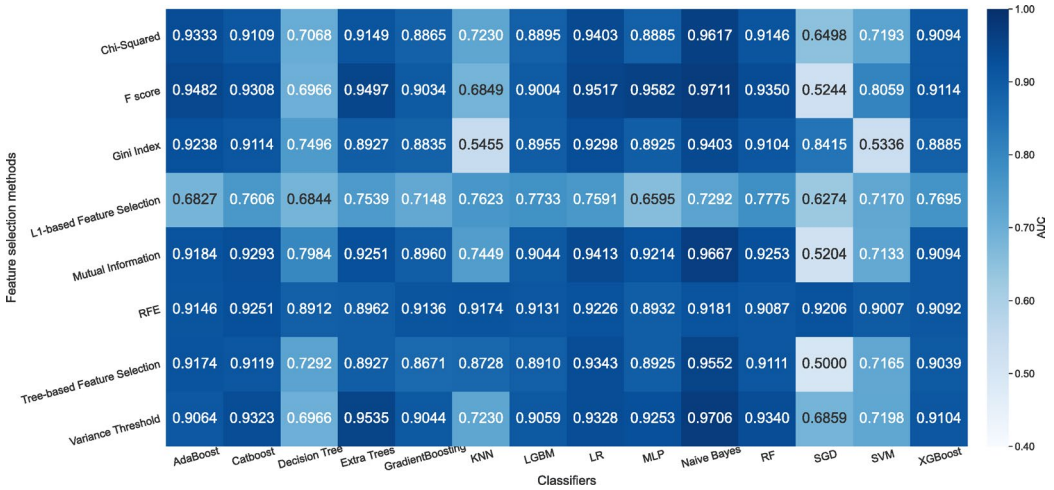


Fig. 4. Distribution of AUC values for the base model on Cleveland dataset.

Predictions and analysis of the base model

This section employs AUC as the core evaluation metric to quantitatively assess the performance of base models combining different feature selection methods with classifiers. Figures 4, 5 and 6 respectively display the AUC value distributions of baseline models across the Cleveland, Framingham, and CVD datasets. All three heatmaps utilize a unified scale (0.4–1.0), enabling direct comparison of model performance differences across datasets.

The result shows that the combination of feature selection methods and classifiers has a significant impact on the AUC performance of the model. Distinct differences in the effects of different combinations. The CCS can generate diverse base models, expand feature dimensions and select efficient combinations, thereby providing high-quality input for stacked ensembles and reducing computational overhead.

Based on the results shown in Figs. 4, 5 and 6, the top 10 base models ranked by AUC and their corresponding AUC values across the three datasets have been summarised. The findings are presented in Table 10.

Predictions results and analysis of the Top 10 base models and the CCS-DWSE model

To comprehensively evaluate the predictive classification performance of the CCS-DWSE model, this paper employs five metrics—Accuracy, Precision, Recall, F1-Score, and AUC—to conduct a multidimensional comparison against the selected Top 10 base models across three datasets (see Tables 11, 12 and 13). The results show that the CCS-DWSE model outperforms the base models across all evaluation metrics: (i) Cleveland dataset: accuracy of 98.32%, AUC of 0.9992, which are 6.1% and 2.76% higher than the best base model, respectively. (ii) Framingham dataset: accuracy of 92.33%, with recall and AUC improved by 15.37% and 20.33%, respectively. (iii) CVD dataset: accuracy improved by 8.44%, AUC by 10.2%, and F1 score by 9.57%.

The results validate that CCS-DWSE, through its ability to capture non-linear features via dynamic fusion of base models, effectively addresses the limitations of single models. It achieves the avoidance of overfitting

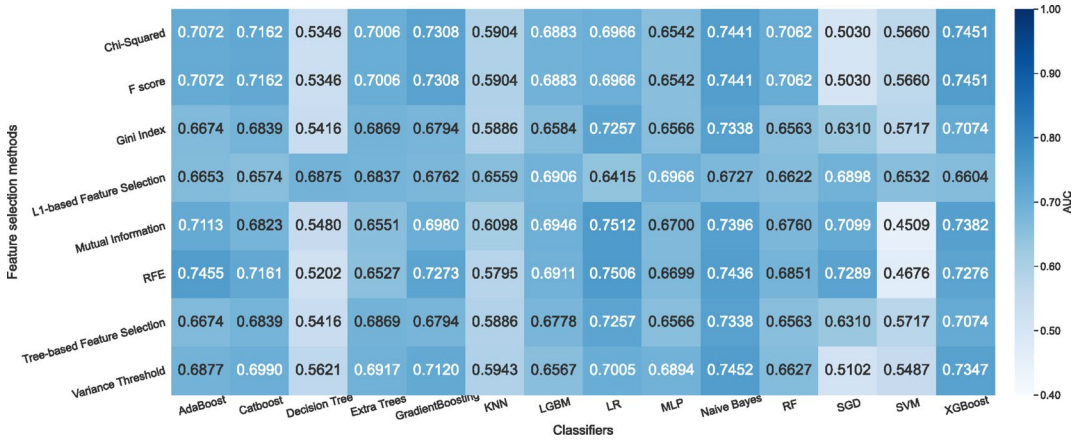


Fig. 5. Distribution of AUC values for the base model on Framingham dataset.

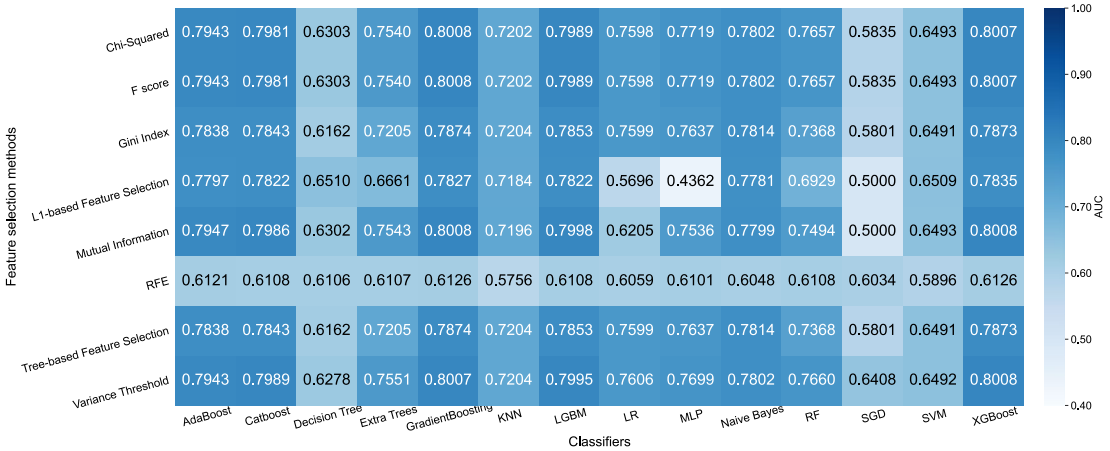


Fig. 6. Distribution of AUC values for the base model on CVD dataset.

on small-sample datasets, relative improvement in minority class recognition on imbalanced datasets, and maintains high accuracy on large-scale sparse datasets. This provides a universal solution for complex and diverse medical prediction tasks.

Figures 7, 8 and 9 show the ROC curves of CCS-DWSE on three datasets. All of the datasets exhibit higher AUC values, indicating that CCS-DWSE effectively reduces the false positive rate while maintaining a high true positive rate. This effect is particularly evident on imbalanced or complex feature distribution datasets, fully demonstrating its stronger generalization competency and robustness.

CCS-DWSE model runtime

To quantitatively validate the algorithm's execution efficiency, this study recorded the actual runtime of the CCS-DWSE model across three datasets under a standardised hardware environment (Intel Core i7-12700 H processor, 32GB RAM) and software environment (Python 3.9). The specific results are presented in Table 14.

SHAP-based interpretability analysis

To improve the interpretability of the model, this paper visualises its importance using bar charts, analyse the linear relationship between features and target variables using correlation coefficient. (i) In the Cleveland dataset (Fig. 10), the correlation coefficients for "exang" and "fbs" (fasting blood sugar) are 0.966 and 0.965, respectively. Their SHAP values also have high absolute values, indicating a significant influence. (ii) In the Framingham dataset (Fig. 11), the correlation coefficients for "totChol" and "cigsPerDay" are 0.579 and 0.571, respectively. The high absolute SHAP values confirm their significant influence. (iii) In the CVD dataset (Fig. 12), the correlation coefficients for "smoke" and "alco" are 0.735 and 0.724, respectively. The high absolute SHAP values also indicate a significant influence.

Guidelines for Primary Prevention of Cardiovascular Diseases in China[32] identifies smoking, dyslipidaemia, and diabetes as major risk factors for cardiovascular diseases (CVDs), and the SHAP analysis results of the CCS-DWSE model are consistent with these guidelines. The SHAP contribution values of indicators such as smoking, alcohol consumption, and blood glucose are relatively high, whereas those of traditional risk factors

Dataset	Number	Model	AUC
Cleveland	1	Mutual Information - Naive bayes	0.9716
	2	F score - Naive bayes	0.9711
	3	Variance Threshold - Naive bayes	0.9706
	4	Chi-Squared - Naive bayes	0.9617
	5	F score - MLP	0.9582
	6	Variance Threshold - Extra Trees	0.9535
	7	F score - Logistic Regression	0.9517
	8	F score- Extra Trees	0.9497
	9	F score -AdaBoost	0.9482
	10	Mutual Information - Extra Trees	0.9221
Framingham	1	RFE - LR	0.7506
	2	RFE - AdaBoost	0.7455
	3	Variance Threshold - Naive bayes	0.7452
	4	F score- XGBoost	0.7451
	5	Chi-Squared - XGBoost	0.7451
	6	F score - Naive Bayes	0.7441
	7	Chi-Squared- Naive Bayes	0.7441
	8	RFE - Naive Bayes	0.7436
	9	Mutual Information-Naive Bayes	0.7430
	10	Variance Threshold -XGBoost	0.7347
CVD	1	Variance Threshold - XGBoost	0.8008
	2	F score - GradientBoost	0.8008
	3	Mutual Information - GradientBoost	0.8008
	4	Chi-Squared - GradientBoost	0.8008
	5	Mutual Information - XGBoost	0.8008
	6	Variance Threshold - GradientBoost	0.8007
	7	F score- XGBoost	0.8007
	8	Chi-Squared - XGBoost	0.8007
	9	Mutual Information -LGBM	0.7998
	10	Variance Threshold- LGBM	0.7995

Table 10. Top 10 base model combinations by AUC across datasets.

Model	Accuracy	Precision	Recall	F1-score	AUC
Mutual Information - Naive bayes	0.9222	0.9474	0.8780	0.9114	0.9716
F score - Naive bayes	0.9222	0.9474	0.8780	0.9114	0.9711
Variance Threshold - Naive bayes	0.9111	0.9459	0.8537	0.8974	0.9706
Chi-Squared - Naive bayes	0.8889	0.9429	0.8049	0.8684	0.9617
F score - MLP	0.8889	0.8605	0.9024	0.8810	0.9582
Variance Threshold - Extra Trees	0.8889	0.8780	0.8780	0.8780	0.9535
F score - Logistic Regression	0.9222	0.9474	0.8780	0.9114	0.9517
F score- Extra Trees	0.9111	0.9459	0.8537	0.8974	0.9497
F score -AdaBoost	0.8556	0.8500	0.8293	0.8395	0.9482
Mutual Information - Extra Trees	0.8667	0.8537	0.8537	0.8537	0.9221
CCS-DWSE	**0.9832**	**0.9853**	**0.9781**	**0.9817**	**0.9992**

Table 11. Cleveland dataset top 10 base models vs. CCS-DWSE prediction results.

including age and blood pressure are relatively low. The specific reasons are as follows: smoking, dyslipidaemia, and abnormal blood glucose are closely associated with pathological processes such as vascular endothelial injury and metabolic disorders, which directly accelerate the occurrence and progression of atherosclerosis. Thus, they are defined as modifiable high-risk factors explicitly specified in the guidelines. In contrast, age is a non-modifiable factor, and blood pressure is an indicator with feasible clinical regulation potential, which accounts for their relatively limited impact in the model.

Model	Accuracy	Precision	Recall	F1-score	AUC
RFE - LR	0.8804	0.6420	0.1670	0.3280	0.7506
RFE - AdaBoost	0.8774	0.4000	0.1670	0.3200	0.7455
Variance Threshold - Naive bayes	0.8480	0.3370	0.2583	0.2925	0.7452
F score- XGBoost	0.8794	0.5455	0.3500	0.3916	0.7451
Chi-Squared - XGBoost	0.8794	0.5455	0.3500	0.3916	0.7451
F score - Naive Bayes	0.8440	0.3191	0.2500	0.2804	0.7441
Chi-Squared- Naive Bayes	0.8440	0.3191	0.2500	0.2804	0.7441
RFE - Naive Bayes	0.8744	0.4000	0.2667	0.2143	0.7436
Mutual Information-Naive Bayes	0.8440	0.3333	0.2833	0.3063	0.7430
Variance Threshold -XGBoost	0.8794	0.5556	0.2417	0.2944	0.7347
CCS-DWSE	**0.9233**	**0.9039**	**0.5037**	**0.6193**	**0.9539**

Table 12. Framingham dataset top 10 base models vs. CCS-DWSE prediction results.

Model	Accuracy	Precision	Recall	F1-score	AUC
Variance Threshold - XGBoost	0.7366	0.7488	0.6969	0.7219	0.8008
F score - GradientBoost	0.7357	0.7461	0.6992	0.7219	0.8008
Mutual Information - GradientBoost	0.7359	0.7457	0.7006	0.7225	0.8008
Chi-Squared - GradientBoost	0.7357	0.7461	0.6992	0.7219	0.8008
Mutual Information - XGBoost	0.7361	0.7487	0.6955	0.7211	0.8008
Variance Threshold - GradientBoost	0.7361	0.746	0.7007	0.7226	0.8007
F score- XGBoost	0.7351	0.7481	0.6934	0.7197	0.8007
Chi-Squared - XGBoost	0.7351	0.7481	0.6934	0.7197	0.8007
Mutual Information -LGBM	0.7361	0.7494	0.6942	0.7207	0.7998
Variance Threshold- LGBM	0.7348	0.7470	0.6947	0.7200	0.7995
CCS-DWSE	**0.8210**	**0.8234**	**0.8119**	**0.8176**	**0.9028**

Table 13. CVD dataset top 10 base models vs. CCS-DWSE prediction results.

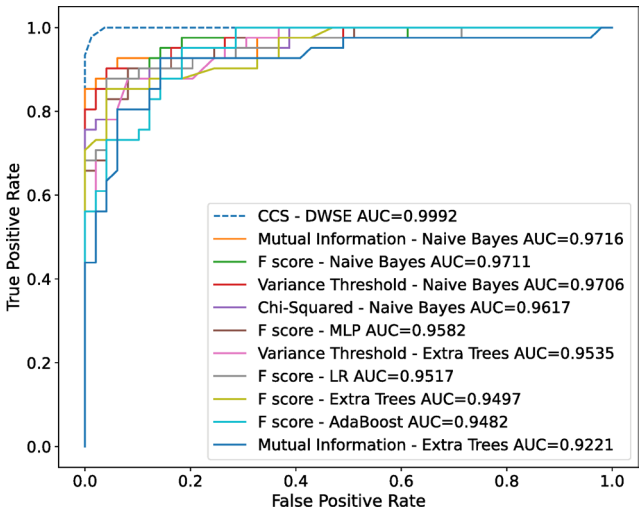


Fig. 7. ROC curve based on the Cleveland dataset.

Comparative experiment

To validate the effectiveness of the CCS-DWSE model, we compared its experimental results on three datasets with those of similar recent studies (see Table 15).

Table 15 shows that the CCS-DWSE model outperforms the comparison model across all metrics. On the Cleveland dataset, its AUC value is 0.9992 (close to 1), demonstrating extremely high predictive accuracy and discriminative ability. Although the accuracy (0.8039) and recall (0.5037) on the Framingham dataset

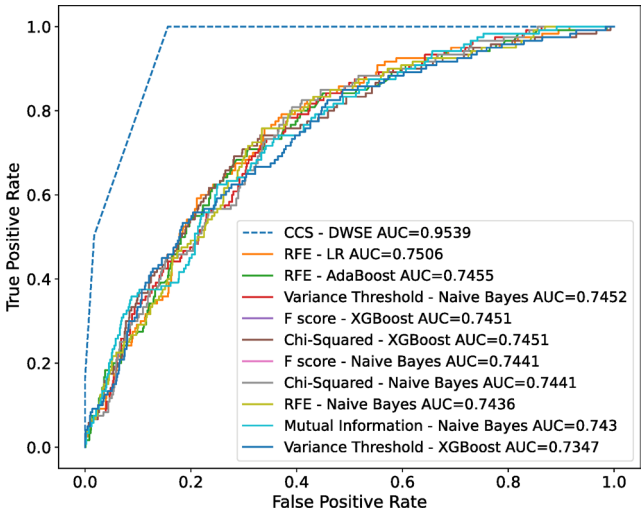


Fig. 8. ROC curve based on the Framingham dataset.

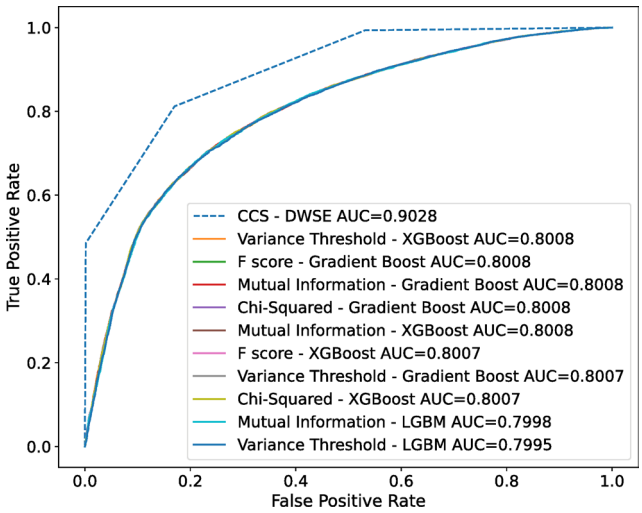


Fig. 9. ROC curve based on the CVD dataset.

Dataset	Average training time (seconds)	Average ensemble time (seconds)	Average total time (seconds)
Cleveland ($N=303$, F = 14)	26.05 ± 2.77	31.00 ± 5.54	57.05 ± 8.31
Framingham ($N=4240$, F = 16)	67.16 ± 3.24	44.03 ± 2.04	101.19 ± 5.28
CVD ($N=70000$, F = 12)	3635.00 ± 5.39	745.73 ± 3.04	4380.73 ± 8.43

Table 14. Training times for CCS-DWSE across different datasets. Note: the average total time in the table is calculated by summing the average training time and average ensemble time. The error term represents the standard deviation across five experiments for each dataset, ensuring the reliability of the results.

are relatively low, its AUC (0.9539) is higher than other models, indicating its strong robustness in handling imbalanced data. The overall performance metrics on the CVD dataset also outperform other models, further validating its broad applicability.

This study used statistical analysis methods, including the Friedman test[33] and Bonferroni-Dunn post hoc test, to verify the statistical significance of the performance differences between CCS-DWSE and the comparison model.

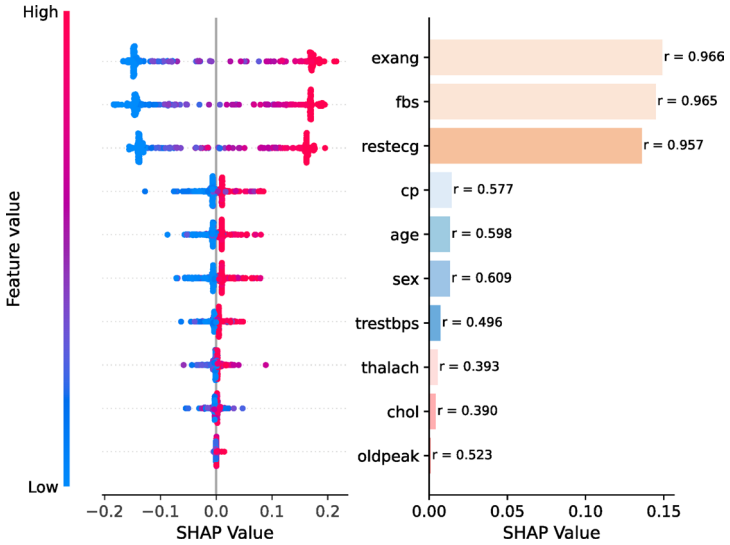


Fig. 10. Feature contribution based on the Cleveland dataset.

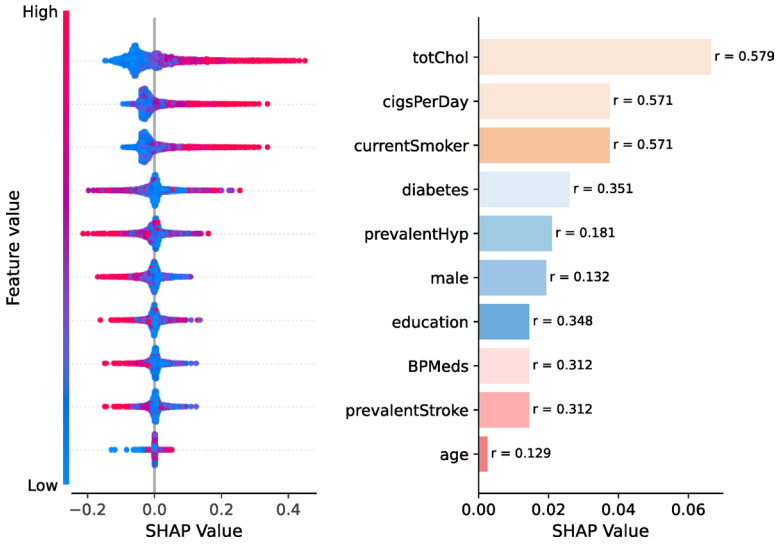


Fig. 11. Feature contribution based on the Framingham dataset.

The Friedman test is a nonparametric method used to evaluate the overall performance differences of k models on n datasets. The Friedman statistic is calculated as follows.

$$\chi_F^2 = \frac{12n}{k(k+1)}\left(\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right) \tag{10}$$

In the formula, R_j is the average rank of the jth model.

$$R_j = \frac{1}{n}\sum_{i=1}^{n} r_i^j \tag{11}$$

In this case, r_i^j denotes the rank of the jth model on the ith dataset. When k and n are large ($k \geq 3$), χ_F^2 approximately follows a chi-square distribution with $k-1$ degrees of freedom. The null hypothesis (H_0) assumes that there is no significant difference in the performance of the models. If the calculated χ_F^2 value exceeds the chi-square distribution's critical value at significance level α, the null hypothesis is rejected, indicating that at least one model's performance differs significantly from the others.

Based on the experimental results, the Friedman statistic χ_F^2 was calculated to be 30.99. At a significance level of $\alpha = 0.05$, the critical value of the F distribution is 6.9443. Since $\chi_F^2 > 6.9443$, the null hypothesis was rejected, indicating a significant difference in performance between the models.

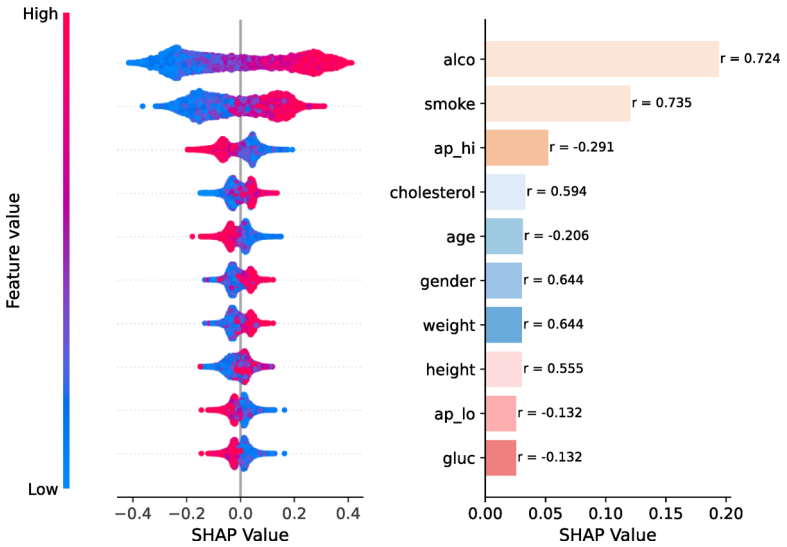


Fig. 12. Feature contribution based on the CVD dataset.

Dataset	Evaluation metrics	Model		
		MCF[11]	XGBLR-IMBODE[23]	CCS-DWSE
Cleveland	Accuracy	0.9562(3)	0.9672(2)	**0.9832(1)**
	Precision	0.9306(3)	0.9655(2)	**0.9853(1)**
	Recall	**0.9781(1)**	0.9655(3)	**0.9781(1)**
	F1-score	0.9537(3)	0.9655(2)	**0.9817(1)**
	AUC	0.9662(2)	—	**0.9992(1)**
	Average ranking	2.4	2.4	**1.2**
Framingham	Accuracy	0.8807(3)	0.9048(2)	**0.9233(1)**
	Precision	**0.8974(1)**	—	0.8039(2)
	Recall	0.2418(2)	—	**0.5037(1)**
	F1-score	0.2798(2)	—	**0.6193(1)**
	AUC	0.7837(2)	—	**0.9539(1)**
	Average ranking	2	2.8	**1.2**
CVD	Accuracy	0.7366(3)	0.7412(2)	**0.8210(1)**
	Precision	0.7581(2)	—	**0.8234(1)**
	Recall	0.6854(2)	—	**0.8119(1)**
	F1-score	0.7199(2)	—	**0.8176(1)**
	AUC	0.8051(2)	—	**0.9028(1)**
	Average ranking	2.2	2.8	**1**

Table 15. Comparison of CCS-DWSE prediction results on different datasets. —Source: ref[23].. The metric was not presented in the original literature.

To further analyse the sources of differences, Bonferroni-Dunn tests were used to calculate critical differences (CD).

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6n}} \tag{12}$$

Here, q_α is the critical value of the studentised range distribution, the calculate d CD value is 1.0564. If the difference in the average rankings of two models exceeds the CD value, the performance difference is statistically significant. Figure 13 shows the statistical analysis results for different models. The results indicate that CCS-DWSE outperforms MCF and XGBLR-IMBODE in terms of performance.

Finally, we compared the performance of the CCS-DWSE model proposed in this paper with the results of several recent studies on the Cleveland dataset (see Table 16). These results indicate that the CCS-DWSE model proposed in this paper outperforms the aforementioned studies in terms of performance.

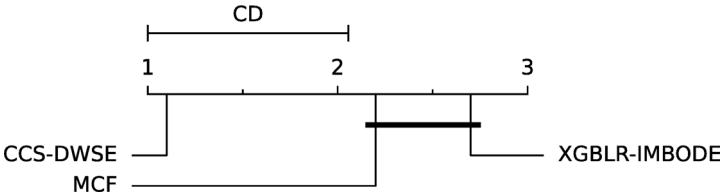


Fig. 13. The Bonferroni-Dunn test results for different models.

Model	Accuracy	Precision	Recall	F1-score
MCF[11]	95.62%	93.06%	**97.81%**	0.9537
RFRS[16]	92.59%	93.33%	91.67%	—
XGBLR-IMBODE[23]	96.72%	96.55%	96.55%	0.9655
Bagging-Fuzzy-GBDT[34]	90.00%	90.00%	94.00%	0.9200
HRFLM[35]	88.40%	90.10%	92.80%	0.9000
Residual GRU+MHSA[36]	89.10%	—	—	0.8590
CCS-DWSE	**98.32%**	**98.56%**	**97.81%**	**0.9817**

Table 16. Multi-model comparison results based on the Cleveland dataset. —Source: ref[16]. and[36]. The metric was not presented in the original literature.

Conclusions

Current models typically employ a single feature selection method and an ensemble strategy with fixed weights, making it difficult to fully explore the complex associations between CVD features and lacking adaptability when faced with different data distributions. To address this problem, the paper proposes CCS-DWSE method. First, the method constructs a heterogeneous model group through the combination of multi-feature selection and multi-classifier to explore the potential associations between features. Then, it selects the top 10 base models with the optimal AUC and adopts a DWSE strategy to dynamically adjust the weights according to the performance of each base model, achieving optimal fusion of prediction results. Finally, SHAP analysis is used to quantify feature contributions. Experimental results show that CCS-DWSE significantly improves prediction performance on the selected public datasets and maintains high robustness in imbalanced data. In addition, SHAP analysis identifies smoking, blood lipids, and blood glucose as key predictors, providing a reference for feature selection in subsequent related studies.

In the data preprocessing stage of this study, the KNNImputer method was used to impute features with a high missing rate, and outliers were standardized. To address the data distribution differences among different datasets, the CCS-DWSE model adaptively assigns differentiated weights based on the real-time prediction performance of base models through a dynamic weight update mechanism, which effectively improves the model's adaptability to data distribution shifts. Its effectiveness and stability have been verified on multiple public datasets. However, the CCS-DWSE model still has certain limitations in practical applications: currently, the model relies on structured features, and the integration of multi-modal data needs to be explored in future research. In response to potential measurement and recording errors during clinical data collection, noise-robust learning mechanisms will be introduced to enhance model stability in subsequent work. This study is conducted based on public datasets, so the generalization ability of the model in real clinical scenarios requires further verification.

Data availability

The data supporting the findings of this study are available via the UCI Machine Learning Repository ([https://archive.ics.uci.edu/]) and the Kaggle platform ([https://www.kaggle.com/]). These datasets originate from the following publicly accessible resources:- Cleveland Dataset: [http://archive.ics.uci.edu/dataset/45/heart+disease](http:/invalid.uri);- Framingham Dataset: [https://www.kaggle.com/datasets/aasheesh200/framingham-heart-st](https:/www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset)-udy-dataset;- CVD Dataset: [https://www.kaggle.com/datasets/mahmudulhaqueshawon/cardiovascular-d](https:/www.kaggle.com/datasets/mahmudulhaqueshawon/cardiovascular-disease/data)-isease/data.

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Author contributions

XBQ designed this study and served as corresponding author. JLG performed the experiments and analysed the data. HQ guided the writing process and secured funding. YS and JJZ contributed to manuscript preparation. All authors reviewed the manuscript and approved the submitted version.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

The cardiovascular dataset employed in this study constitutes publicly available data. Consequently, ethical approval was not required, and the research adhered to relevant data usage and ethical guidelines.

Additional information

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